

2c. Content of modules 1 to 6

Learners are expected to be able to demonstrate and apply their knowledge and understanding of all the learning outcomes in this specification. The learning outcomes in modules 2 – 6 are split into four categories, labelled **(a)**, **(b)**, **(c)**, and **(d)**, to order the specification in a helpful way for planning teaching programmes and to help identify the knowledge required of learners taking the course.

The headings for these categories are:

- (a)** Describe and explain
- (b)** Make appropriate use of
- (c)** Make calculations and estimates involving
- (d)** Demonstrate and apply knowledge and understanding of the following practical activities.

Module 1: Development of practical skills in physics

Physics is a practical subject and the development of practical skills is fundamental to understanding the nature of physics. Physics B gives learners many opportunities to develop the fundamental skills

needed to collect and analyse empirical data. Skills in planning, implementing, analysing and evaluating, as outlined in **1.1**, will be assessed in the written papers.

1.1 Practical skills assessed in a written examination

Practical skills are embedded throughout the content of this specification.

Learners will be required to develop a range of practical skills throughout their course in preparation for the written examinations.

1.1.1 Planning

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) experimental design, including to solve problems set in a practical context	Including selection of suitable apparatus, equipment and techniques for the proposed experiment. Learners should be able to apply scientific knowledge based on the content of the specification to the practical context. HSW3
(b) identification of variables that must be controlled, where appropriate	
(c) evaluation that an experimental method is appropriate to meet the expected outcomes.	HSW6